Sean Hyde
Active Secret Clearance (2026)
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Professional Summary

Senior Computer Science student at Rensselaer Polytechnic Institute, graduating December 2026, with professional experience developing LLM-driven systems, AI agents, and full-stack applications. Holds an active Secret security clearance. Seeking an entry-level software engineering or AI/ML engineering position to apply my technical experience while continuing to grow as an engineer.

Education

- **Rensselaer Polytechnic Institute** — Troy, NY
  Bachelor of Science in Computer Science — *Aug. 2024 – Dec. 2026*
- **SUNY Polytechnic Institute** — Utica, NY
  Computer Science Coursework — *Aug. 2022 – April. 2024*

Research & Industry Experience

- **Griffiss Institute – AFRL Contract** — Rome, NY
  ATLAS — AI Training Simulation Platform (Sole Developer) — *May 2025 – Present*
  - **Simulation engine**: Architected a headless Rust simulation engine (Bevy ECS) for LLM-agent training scenarios. Designed a shared `scenario-core` crate and contingency pools that adapt scenarios to trainee decisions and ground information within the scenario.
  - **LLM agent service**: Built a Python agent service on a Redis priority queue with multi-turn tool calling, per-room locking, and ordered delivery. Agents act through a world-model proxy and a 23-tool MCP server, backed by hybrid memory (Qdrant + sentence-transformers) and EMA-weighted LLM-as-a-judge scoring.
  - **Instructor dashboard**: Tauri app (React 19 + TypeScript, Rust) for scenario authoring, deployment, and live instruction, with WebSocket streaming, geospatial track playback, and LLM scenario generation with structured output validation.
  - **Deployment and performance**: 6-service Docker Compose stack built for offline delivery. Cut message latency from ∼12s to sub-second and eliminated sustained 100% CPU usage.
- **Griffiss Institute** — Rome, NY
  Research Intern – AI/LLM Research — *May 2025 – Aug 2025*
  - **Large language model research**: Supported ML R&D on LLMs, designing experiments for training, optimization, and information retrieval. Reviewed LLM/RAG literature and analyzed results for accuracy, latency, and robustness.
  - **Tool- and agent-based systems**: Designed tool- and agent-based LLM workflows using MCP-style orchestration and agent-to-agent coordination for multi-step pipelines and external tool interaction, measuring task completion and robustness.
  - **Retrieval-Augmented Generation (RAG)**: Implemented RAG pipelines spanning ingestion, chunking, embedding, retrieval, and generation; ran experiments on retrieval parameters and prompt strategies to evaluate quality, latency, and correctness trade-offs.
  - **Model training with Unsloth**: Fine-tuned LLMs on domain-specific datasets using Unsloth.
  - **HPC and containerized workflows**: Packaged experiments in Docker and deployed them on an HPC cluster, managing jobs, resources, and experiment tracking for large-scale training and evaluation runs.
  - **Evaluation and documentation**: Built evaluation harnesses, including LLM-as-a-judge, to evaluate model outputs at scale; documented best practices for future researchers and engineers.

Accomplishments & Leadership Experience

- **1st Place – RI Intern and Information Institute Poster Session (2025)**: Awarded first place among 80+ participating interns at Griffiss Institute for developing a research poster and delivering a technical pitch on my internship project to Griffiss Institute and AFRL employees.
- **3rd Place – AFRL Challenge Competition (2022)**: Earned third place alongside a partner in a week-long competition hosted at Griffiss Institute. Optimized a machine learning model to classify images of humans, dogs, and cats, presenting our approach and results to Griffiss Institute and AFRL leadership.
- **Team Lead – HackRPI Discord Bot Development (2025)**: Led a small development team responsible for building a Discord bot used to communicate with HackRPI participants. Delegated development tasks and authored technical onboarding documentation covering environment setup, bot configuration, and development workflows.

Projects

- **Self-Hosted Homelab** — 2026
  Personal Arch Linux server – infrastructure, monitoring & DR — *Jun 2026 – Present*
  - **Services and networking**: Maintain a Docker Compose stack including Gitea, Vaultwarden, and FreshRSS, with Watchtower updates and Dozzle logs; configure AdGuard Home, Unbound DNS, and Tailscale remote access.
  - **Monitoring and recovery**: Developed cron-based Bash health checks for disk health, temperature, updates, and container status with Discord alerts; use restic for backups.
- **EverQuest Item Finder Bot** — 2024
  Discord bot for natural-language item lookup
  - **Natural-language search**: Used the OpenAI API to translate user natural language inputs into structured PostgreSQL queries through prompt templates and parsing logic; exposed item lookup through a Discord bot and Flask web interface.

Skills

- **Languages: Python**, Rust, TypeScript, JavaScript, Java, C, C++, HTML & CSS; **ML/AI:** vLLM, ollama, LM Studio, NumPy, HuggingFace Transformers, Unsloth, Qdrant, sentence-transformers; **LLM Tooling:** MCP, RAG, OpenAI function calling, multi-turn tool loops, LLM-as-a-judge evaluation; **Full-Stack:** React, Tauri, Tailwind CSS, Vite; **DevOps/Data:** Docker, Docker Compose, Git/GitHub/GitLab, MongoDB, PostgreSQL, Redis, WebSocket/DDP; **AI Tools:** OpenCode, Claude Code, Hermes